# Student Club Registration System

Low-code application built on Zoho Creator

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CB23F35 - Low-Code Application Development using Zoho Creator

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# Agenda

1. Introduction and problem statement
2. Objectives and scope
3. System study and proposed system
4. Requirements and use cases
5. System design: architecture, data model, access control
6. Implementation: forms, reports, dashboards, Deluge, workflows
7. Testing, deployment and security
8. Results, conclusion and future work

# Introduction

- Student clubs encourage technical learning, creativity, leadership and collaboration on campus.
- Membership and event participation are often managed through **paper forms, spreadsheets and messaging apps**.
- Information is scattered, so coordinators verify details, check eligibility and communicate decisions by hand.
- A **centralised digital system** makes registration simpler, faster and more consistent.
- This project delivers one such system on the **Zoho Creator** low-code platform.

# Problem Statement

### Scattered information

Student and club records live in separate files or messages.

### Duplicate applications

The same student may apply to a club more than once.

### Slow approval

Manual review; applicants do not know the outcome.

### Limited visibility

Club strength, vacancies and pending requests are hard to track.

### Event overhead

Attendance and participant lists are prepared separately.

### Weak traceability

Status changes are not recorded consistently.

# Objectives and Scope

## Objectives

- Central data model for students, clubs, memberships, events
- Simple interface to browse clubs and request membership
- Automated duplicate checks, validation, approvals, notifications
- Reports and dashboards for membership and event participation
- Role-based access with an audit trail

## Scope

### IN SCOPE

- Student profile, club catalogue, membership requests, approval / rejection, events, event registration, status tracking, reports, dashboards, access control

### OUT OF SCOPE (FUTURE)

- Fee collection, biometric attendance, university ERP integration, automated certificates

# Manual vs Low-Code Registration

| Criterion | Manual / Spreadsheet | Proposed Low-Code System |
|---|---|---|
| Data capture | Separate forms and files | Centralised validated forms |
| Duplicate control | Manual checking | Workflow-based duplicate check |
| Approval | Messages or email | Structured status and review |
| Status visibility | Follow-up required | Reports and dashboards |
| Event participants | Separate lists | Linked event registrations |
| Access control | File sharing | Role and record permissions |
| Auditability | Limited history | Action log and timestamps |

# Proposed System and Low-Code Justification

- **One application** for students, club coordinators and administrators.
- Students maintain profiles, explore clubs, submit membership and event requests.
- Coordinators review requests and maintain club and event details.
- Administrators manage master data, users and access.
- Zoho Creator provides visual form/report builders, **lookup relationships**, workflows, **Deluge scripting**, role profiles and responsive layouts.
- Development effort goes to registration rules, approval logic and usability instead of boilerplate code.

# Requirements: Roles and Functions

| Role | Responsibilities |
|---|---|
| Administrator | Manage users, clubs, master data, permissions and audit records |
| Student | Maintain profile, browse clubs, apply and track status |
| Club Coordinator | Update club info, review requests, manage events and member lists |

- **Key functional requirements:** student profile, club browsing, membership request, duplicate prevention, approve/reject with remarks, status notifications, event creation and registration, dashboards, audit and RBAC.
- **Non-functional:** usability, performance, security, reliability, maintainability, auditability.
- **Platform:** Zoho Creator + Deluge; Chrome/Edge on laptop, desktop or smartphone.

# Use Case Summary

| Use Case | Actor | Description |
|---|---|---|
| UC-01 Browse clubs | Student | View active clubs, eligibility and capacity |
| UC-02 Apply for membership | Student | Submit a request and receive a request ID |
| UC-03 Review application | Coordinator | Approve/reject and enter remarks |
| UC-04 Register for event | Student | Select event and submit registration |
| UC-05 Manage club/event | Coordinator | Maintain details, capacity and schedule |
| UC-06 Administer system | Administrator | Manage accounts, roles and master records |
| UC-07 Monitor activity | Admin / Coordinator | View reports and dashboards |

# System Architecture

| | |
|---|---|
| **Presentation** | Student UI, Coordinator UI, Admin dashboard, reports |
| **Application & Logic** | Form workflows, duplicate validation, approval process, Deluge scripts, notifications |
| **Data & Storage** | Creator forms, lookup relationships, registration records, audit log |

**Security**

Authentication, role profiles and record-level criteria apply across all three layers.

# Data Model

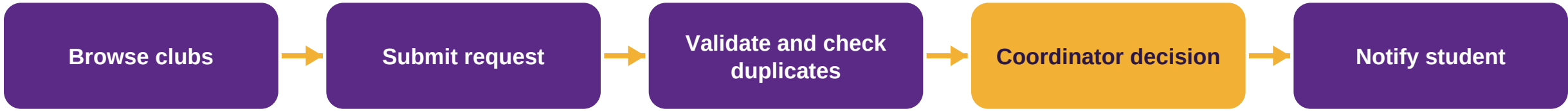
Event Registration also links to Student Profile (a student has many event registrations).

Club Management also links to Event Management (a club has many events).

Admin Log records key actions: action, entity, record ID, actor, timestamp.

# Membership Form and Workflow

| Field | Type | Validation / Remarks |
|---|---|---|
| Request ID | Auto Number | Generated automatically |
| Student | Lookup | Required; linked to Student Profile |
| Club | Lookup | Required; only active clubs |
| Application Date | Date | Default current date |
| Status | Drop-down | Pending, Approved, Rejected, Withdrawn |
| Coordinator Remarks / Decision Date | Multi-line / Date-Time | Filled during review |



# Access Control Design

| Component | Admin | Student | Coordinator |
|---|---|---|---|
| Student Profile | AEVD | Own: V/E | V (assigned club context) |
| Club Management | AEVD | V | V/E (own club) |
| Membership Registration | AEVD | A/V (own requests) | V/E (assigned club) |
| Event Management | AEVD | V | AE (own club) |
| Event Registration | AEVD | A/V (own) | V (own events) |
| Admin Log | V | - | - |
| Dashboard | V | Student dashboard | Coordinator dashboard |

*A = Add, E = Edit, V = View, D = Delete.  Least-privilege profiles with record-level criteria.*

# Implementation: Forms, Reports, Dashboards

## Forms

- Student Profile
- Club Management
- Membership Registration
- Event Management
- Event Registration
- Admin Log

## Reports

- Student List
- Active Clubs
- Pending Membership Requests
- Club Members
- Event Registrations
- Admin Log Report

## Dashboards

- Student: clubs, my memberships, upcoming events
- Coordinator: pending requests, members, capacity
- Admin: totals for students, clubs, requests, event registrations

# Implementation: Deluge Scripts and Workflows

| Script | Trigger | Purpose |
|---|---|---|
| DS-01 Duplicate Check | Form validation | Block duplicate pending/approved requests |
| DS-02 Request Initialization | On submission | Set Pending status and timestamp |
| DS-03 Decision Update | Coordinator action | Update status, decision date, remarks |
| DS-04 Student Notification | Status change | Notify student of decision |
| DS-05 Event Capacity Check | Event registration | Reject if capacity reached |
| DS-06 Audit Entry | Important action | Write actor, action, record, timestamp |

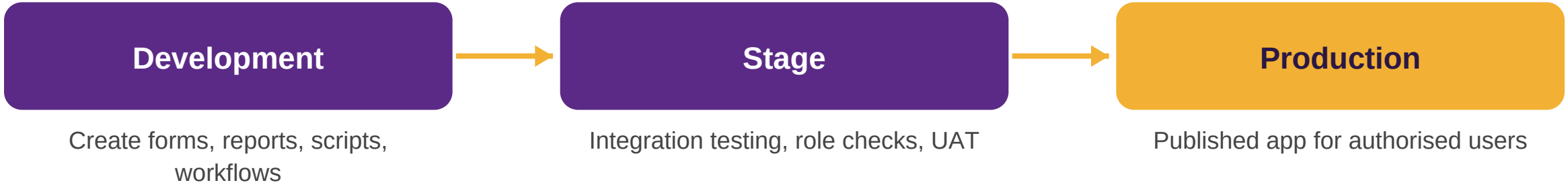
```
existing = Membership_Registration[Student == input.Student & Club == input.Club
                  & (Status == "Pending" || Status == "Approved")];
if(existing.count() > 0) { alert "You already have an active or pending request for this club.";
                           cancel submit; }
```

# Testing and Debugging

| ID | Test | Expected Result |
|---|---|---|
| TC-02 | Incomplete profile | Validation message; save blocked |
| TC-03 | Membership request | Saved as Pending |
| TC-04 | Duplicate request | Duplicate blocked |
| TC-05 | Coordinator approval | Approved; student notified |
| TC-08 | Event capacity full | Registration blocked |
| TC-09 | Open another student's record | Access denied |
| TC-10 | Change request status | Actor and timestamp recorded |

- Four levels: unit, integration, role-based and user acceptance testing in the stage environment.
- Defects fixed: missing status filter in duplicate check, overly broad coordinator criteria, capacity race condition, empty email recipient.

# Deployment, Security and Governance




- Role-based permission profiles and record-level criteria
- Authenticated HTTPS access and platform-managed storage
- Input validation and controlled status transitions
- Audit logging for approvals and admin changes
- Periodic review of accounts and roles; retention per institutional policy

# Results and Discussion

| Objective | Outcome |
| --- | --- |
| Central data model | Linked student, club, membership and event records |
| Simple registration | Club browsing and request form |
| Automated review | Pending queue, decision status, notification |
| Reporting | Membership and event dashboards |
| Secure access | Role profiles and record-level restrictions |
| Traceability | Audit entries for key status changes |

- **Limitations:** edition limits, custom UI may need widgets, concurrent capacity checks need care, integrations depend on API approval.

# Conclusion and Future Enhancements

## Conclusion

- Brings profiles, clubs, requests, approvals and events into one application
- Validations, workflows and reports reduce fragmented data and manual follow-up
- Low-code makes the solution practical to build, test and maintain

## Future Enhancements

- Institutional student information system integration
- Calendar sync and automated event reminders
- QR-based event check-in and attendance
- Certificate generation for participants
- Interest-based club recommendations
- Waitlists and vacancy allocation
- Advanced participation analytics

# Thank You

Questions?

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